

PROFESSIONAL ETHICS FOR COMPUTING

Why We Need a Code of Ethics

Real incidents, the ACM Code of Ethics, cross-disciplinary parallels, and the research-integrity policies built on top of it

Case Studies

ACM Code

Cross-Discipline

Research Integrity

August 2026

Seven Incidents That Actually Happened

- These seven incidents span several layers at which ethical issues arose.
- Every one produced a real, public consequence: a resignation, a recall, a \$2.5 billion settlement, a Supreme Court petition.
- **None of them had a clean answer at the time — that's the point. Each slide closes with a discussion question that's still open.**

Who Is Responsible for What a Platform Hosts?

- August 2017: days after the Charlottesville rally, where a car attack killed counter-protester Heather Heyer, the neo-Nazi site the Daily Stormer published an article mocking her death. GoDaddy and Google had already dropped the site; Cloudflare's DDoS protection was its last major line of defense, after years of insisting it stays "content neutral" no matter how objectionable a client's content is.
- Cloudflare CEO Matthew Prince terminated the account unilaterally, without board discussion, admitting in an internal memo that the decision was arbitrary and driven by personal frustration rather than any change in policy. The site went offline within hours, taken down by attackers the moment protection stopped.
- Prince then published a public blog post warning that a system where one CEO can make this call on a whim is dangerous — even though he didn't reverse the decision.

"I woke up in a bad mood and decided someone shouldn't be allowed on the Internet."

— Matthew Prince, CEO, Cloudflare (internal memo, Aug. 16, 2017)

Discussion: Cloudflare had a stated neutrality policy for years, then broke it in one unilateral, admittedly arbitrary decision. Is inconsistent enforcement worse than having no policy at all?

Whose Contract Is It, Really?

- 2017: Google Cloud quietly signed a contract with the Pentagon's “Project Maven” to apply machine-learning object-recognition to analyze drone surveillance footage — reportedly worth around \$9 million, small for Google but symbolically significant.
- When employees discovered the contract, an internal letter to CEO Sundar Pichai demanding cancellation gathered signatures from more than 3,000 employees; roughly a dozen engineers resigned in protest, some explicitly citing the contract as their reason for leaving.
- In June 2018, Google announced it would not renew the Maven contract once it expired, and published a public set of AI Principles — including an explicit commitment not to design or deploy AI for weapons.
- The episode became the reference case for a broader question the industry still hasn't settled: what happens when a company's own engineers, not regulators or the public, are the ones who object to a contract?

Discussion: Does an engineer's ethical responsibility end at writing correct, working code, or extend to what a client ultimately does with it?

When the Design Brief Is the Ethical Violation

- For years, Amazon enrolled customers into Prime through checkout flows the FTC later alleged were deliberately confusing — while making cancellation a four-page, six-click, fifteen-option maze employees privately nicknamed the “Iliad Flow,” after Homer’s decade-long war. Internal documents showed employees describing Prime enrollment as “a bit of a shady world.”
- Leadership allegedly slowed or rejected proposed fixes to the cancellation flow out of concern they would hurt revenue. The FTC sued in June 2023; in September 2025, days before trial, Amazon agreed to a \$2.5 billion settlement — a \$1 billion civil penalty (the largest the FTC has ever obtained for this type of violation) plus \$1.5 billion in refunds to roughly 35 million customers.
- The settlement also imposes a 10-year injunction against similar dark patterns and requires an independent monitor — turning what had been treated as an internal design decision into an externally supervised legal obligation.

“an unspoken cancer”

— Internal Amazon communication describing unwanted subscriptions, cited in FTC litigation

Discussion: “That’s what the client asked for” is common in industry, but it’s a description of a request, not an ethical defense. What should an engineer actually do when asked to build something manipulative?

Accessibility Isn't Optional Just Because It's Hard

- Guillermo Robles, who is blind and uses screen-reading software, tried repeatedly to order from Domino's website and mobile app and found both incompatible with his screen reader — he could not complete an order through either channel.
- He sued in 2016 under the Americans with Disabilities Act (1990). Domino's argued the ADA shouldn't apply to websites at all, since the Department of Justice had never issued specific technical accessibility standards for the web.
- The Ninth Circuit disagreed, ruling that a website or app tied to a physical place of business must be accessible under the ADA regardless of the absence of formal technical standards. Domino's appealed to the Supreme Court, backed by the US Chamber of Commerce.
- In October 2019, the Supreme Court declined to hear the case, letting the Ninth Circuit's ruling stand nationwide. The case proceeded to trial, and the parties ultimately settled in June 2022 — six years after Robles first tried to order a pizza online.

Discussion: Accessibility fixes are almost always cheaper the earlier they're made. Why do they so reliably get deprioritized until a lawsuit forces the issue?

A Filter That “Works” Can Still Be Broken

- The US Children's Internet Protection Act (CIPA, 2000) requires any school or library receiving federal E-rate funding to install filters blocking obscene material, child sexual abuse material, and content “harmful to minors.”
- In practice, filtering vendors implement this with broad keyword and category blocking — flagging anything tagged with terms like “LGBTQ” or “sexuality” as a category, regardless of whether the actual content is explicit.
- Studies by the American Library Association and the Electronic Frontier Foundation have repeatedly found this over-blocks legitimate, non-explicit resources: sex-education material, suicide-prevention hotlines, and LGBTQ+ support and history sites.
- The asymmetry driving this: under-blocking risks a school's federal funding and legal liability, while over-blocking carries no penalty at all — so vendors default to blocking broadly, and the harm falls hardest on students who may have no other private way to reach these resources.

Discussion: A content filter's accuracy metric usually only measures what it correctly blocks. How should a team be required to measure the harm from what it wrongly blocks?

Security Debt in a Device That Keeps You Alive

- 2016: cybersecurity firm MedSec discovered serious vulnerabilities in St. Jude Medical's implantable pacemakers and defibrillators — within radio range, an attacker could potentially deplete the battery or deliver inappropriate pacing signals.
- Instead of privately disclosing the flaw to St. Jude first, MedSec partnered with investment firm Muddy Waters Capital, which publicly announced the vulnerabilities while simultaneously short-selling St. Jude's stock — profiting directly from the resulting price drop.
- St. Jude initially denied the findings and sued MedSec and Muddy Waters for defamation and securities fraud.
- In August 2017, the FDA confirmed the vulnerabilities were real and recalled roughly 465,000 devices, requiring an in-clinic firmware update — not a remote patch — to fix them.

Discussion: MedSec profited financially from the timing of its disclosure instead of quietly notifying St. Jude first. Does profiting from a disclosure change whether the disclosure itself was the right thing to do?

When “Intense Culture” Is a Euphemism

- November 2015: Susan Fowler joined Uber as a site reliability engineer. On her first official day on a new team, her manager propositioned her over company chat, saying he was looking for partners at work. She reported it to HR with screenshots.
- HR told her this was his “first offense” and that he was “a high performer,” so no serious action would be taken — despite Fowler later learning multiple other women had already reported the same manager. Over the following year she was denied transfers and told a second complaint could get her fired for “poor performance.”
- She published her account on February 19, 2017, after leaving for Stripe. The post went viral, #DeleteUber trended, the board hired former US Attorney General Eric Holder to investigate, over 20 employees were fired, and CEO Travis Kalanick resigned that June.

“he was a high performer”

— HR's characterization of the manager, as reported by Susan Fowler

Discussion: Fowler followed the official reporting process exactly as designed, and it protected the harasser instead of her. What does that imply about relying on internal HR processes alone?

Where the Code Comes From

- First written in 1992; substantially rewritten by the Code 2018 Task Force after two years of public drafts and member feedback.
- **The Preamble states the single organizing rule explicitly: when principles conflict, computing professionals should defer to the public good as the paramount consideration.**
- It is explicitly not an algorithm: “The Code is not an algorithm for solving ethical problems; rather it serves as a basis for ethical decision-making.” Different principles will have different relevance to different situations — judgment is still required.

General Ethical Principles (1.1–1.7)

1.1 Contribute to society and human well-being

1.2 Avoid harm

1.3 Be honest and trustworthy

1.4 Be fair and take action not to discriminate

1.5 Respect the work required to produce new ideas, inventions, and creative works

1.6 Respect privacy

1.7 Honor confidentiality

Professional Responsibilities (2.1–2.9)

- 2.1 Strive to achieve high quality in both the processes and products of professional work
- 2.2 Maintain high standards of professional competence, conduct, and ethical practice
- 2.3 Know and respect existing rules pertaining to professional work
- 2.4 Accept and provide appropriate professional review
- 2.5 Give comprehensive and thorough evaluations of computer systems, including risk analysis
- 2.6 Perform work only in areas of competence
- 2.7 Foster public awareness and understanding of computing, related technologies, and their consequences
- 2.8 Access computing and communication resources only when authorized
- 2.9 Design and implement systems that are robustly and usably secure

2.5 and 2.9 are examined in their own words on the next slide.

2.5 and 2.9, in the Code's Own Words

2.5 — Thorough Evaluation

“...comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks.”

- This is the direct professional obligation behind every “we didn't think it would be used that way” excuse — the Code says thinking it through in advance is the job itself, not an optional extra.

2.9 — Robust, Secure Design

“Design and implement systems that are robustly and usably secure.”

- The technical twin of 2.5: robustness isn't a nice-to-have feature request, it's a named ethical obligation.

Leadership Principles & Compliance

Section 3 — Leadership Principles

- Applies to anyone in a leadership role — in a company, a research group, or a volunteer capacity, e.g. within ACM itself.
- 3.1 — ensure that the public good is a central concern during all professional work undertaken on behalf of an organization.
- 3.4 — manage personnel and resources to enhance the quality of working life — directly relevant to Jean, the manager in the Uber-adjacent scenario.
- 3.7 — recognize and take special care of systems that become integrated into society's infrastructure.

Section 4 — Compliance

- 4.1 — uphold, promote, and respect the principles of the Code.
- 4.2 — treat violations of the Code as inconsistent with membership in ACM.
- This is the Code's only real enforcement teeth — professional standing, not a legal penalty.
- **Compare this to how thin GDPR/DPDP enforcement was until real fines started landing — soft norms vs. hard law is a spectrum, not a binary.**

The Hippocratic Oath, Ancient and Modern

- The Hippocratic Oath (~5th century BCE) is the oldest surviving professional ethics code in continuous use. The classical translation includes the promise to act for the benefit of the patient and to avoid harm and injustice — the actual origin of the harm-avoidance principle.

“...I will keep them from harm and injustice.” — Hippocratic Oath, classical translation

- Modern medical ethics formalizes this into four principles (Beauchamp & Childress) every hospital ethics board still uses: Beneficence, Non-maleficence, Autonomy, Justice. Mapping to ACM: 1.1, 1.2, 1.6/1.7, and 1.4 respectively — medicine got there first.

The Russell–Einstein Manifesto (1955)

- Signed by 11 eminent scientists and intellectuals — including Bertrand Russell and Albert Einstein, who signed just days before his death — in the aftermath of hydrogen bomb tests that showed nuclear weapons could now threaten human civilization itself, not just a single city.
- It did not ask governments to trust scientists less — it asked scientists to publicly take responsibility for what their own field had made possible, and to speak plainly to the public about the actual risk, independent of national politics.

“Remember your humanity, and forget the rest.” — closing line of the Russell–Einstein Manifesto

- **Its direct legacy: the Pugwash Conferences on Science and World Affairs, an ongoing series of meetings between scientists across Cold War lines that contributed to arms-control treaties — and won the Nobel Peace Prize in 1995.**
- The parallel for computing: AI researchers today face a structurally similar question — what do you do when your own field builds something whose worst-case consequences are larger than any one company or country?

Asilomar, Twice: Biology, Then AI

Asilomar Conference on Recombinant DNA (1975)

- Molecular biologists, alarmed by their own new gene-splicing techniques, voluntarily paused their research and convened at Asilomar, California.
- They wrote their own safety guidelines — containment levels matched to experiment risk — before any government agency required it.

“...no longer avoid final responsibility for where his research is taking him.” — Paul Berg, Asilomar organizer

Asilomar AI Principles (2017)

- The Future of Life Institute convened AI researchers at the same Asilomar site, deliberately invoking the 1975 precedent, to draft 23 principles for beneficial AI.
- Signed by thousands of researchers, including many building the frontier models discussed in Class 5.

“The goal of AI research should be to create not undirected intelligence, but beneficial intelligence.” — Asilomar AI Principles

The Pattern That Repeats

Field	Code / Document	Provoked by...	Institutional response
Medicine	Hippocratic Oath / 4 Principles	Malpractice, patient harm	Ethics boards, licensing
Physics	Russell–Einstein Manifesto	The atomic and hydrogen bomb	Pugwash, arms control advocacy
Biology	Asilomar (1975)	Recombinant DNA / genetic engineering	Self-imposed containment levels
Computing	ACM Code + Asilomar AI (2017)	Data breaches, algorithmic harm, AI risk	ACM Code, AI safety research

Every mature field converges on the same handful of ideas — and in most of these cases, self-regulation came before external regulation forced it.

Plagiarism: More Categories Than You'd Think

- **Direct / verbatim plagiarism** — copying another's text or ideas without attribution. The obvious case, and the least common one ACM actually has to adjudicate.
- **Paraphrasing plagiarism** — lightly rewording someone else's argument or structure without citation. Changing the words doesn't change whose idea it is — ACM treats close paraphrasing the same as direct copying if the source isn't credited.
- **Self-plagiarism / redundant (“salami-slicing”) publication** — submitting your own previously published work as new, or splitting one study into the minimum publishable units to inflate a CV, without disclosure. Treated as a real violation, not a victimless shortcut — it pollutes the literature and double-counts a result.
- **Mosaic plagiarism** — stitching together phrases and sentences from multiple uncited sources into a single new-looking passage. Often unintentional, but the policy doesn't require intent to find a violation.
- **Collusion** — undisclosed coordination between authors and reviewers, or between competing submissions, to gain an unfair advantage.
- ACM applies a graduated response based on severity and intent — from a correction, to retraction, to a multi-year ban on submitting to ACM venues.

ACM's Authorship Policy

- To qualify as an author, a person must have made a significant intellectual contribution to the work — not just supplied funding, data, or supervision alone.
- Author order should reflect contribution, agreed upon by all authors before submission — not seniority or department politics.
- A corresponding author takes responsibility for the accuracy of the full author list and for representing all co-authors accurately during review.
- **The problem this doesn't solve: “significant intellectual contribution” and “order reflects contribution” are both famously contested in practice — which is exactly why the CRediT taxonomy on the next slide exists.**

The CRediT Taxonomy: All 14 Roles

- CRediT (Contributor Roles Taxonomy, Elsevier/CASRAI/NISO) replaces one ordered author list with 14 standardized, non-exclusive roles. Every author is tagged with every role they actually performed, published alongside the paper.

Conceptualization	Ideas; formulating research goals and aims	Resources	Providing materials, samples, or computing resources
Data curation	Managing, annotating, and maintaining research data	Software	Programming and algorithm implementation
Formal analysis	Statistical or mathematical analysis of study data	Supervision	Oversight and mentorship for the research team
Funding acquisition	Acquiring the financial support for the project	Validation	Verifying reproducibility of results
Investigation	Conducting the research and data/evidence collection	Visualization	Preparing data presentation and visualization
Methodology	Developing or designing methodology and models	Writing – original draft	Preparing the initial written draft
Project administration	Managing and coordinating research activity planning	Writing – review & editing	Critical review, commentary, and revision

Conflict of Interest

- A conflict of interest exists whenever a personal, financial, or professional relationship could reasonably be seen to bias a decision — not just when it actually does. The Code cares about the appearance of bias as much as bias itself, because trust in the process is the thing being protected.
- Covers reviewers evaluating a paper from their own advisor, student, close collaborator, or a direct competitor; editors handling a submission from their own institution; and financial relationships with a company whose product is being evaluated.
- **The mechanism is disclosure, not prohibition: a conflict doesn't automatically disqualify someone — undisclosed conflicts are the actual violation, because they deny everyone else the chance to weigh the decision accordingly.**

Peer Review

- Confidentiality: a submission under review is privileged information. A reviewer may not share it, discuss its ideas as their own, or use it to inform their own competing work before it's public — the formal version of “no scooping.”
- Timeliness and fairness: reviewers commit to evaluating work on its merits, within a reasonable timeframe, and to declining an assignment outright if they have a conflict of interest they can't set aside.
- **Single-blind vs. double-blind: reveals a real design trade-off — double-blind (reviewers don't know authors) reduces reputation bias but is imperfect once venues, writing style, or self-citations reveal identity anyway; ACM venues vary in which model they use.**

Reproducibility

- CACM has run repeated pieces on empirical computer science's own reproducibility problem — results that can't be independently re-derived because code, data, or environment weren't preserved, not because the finding was fraudulent.
- ACM's response: an artifact-evaluation and badging system — papers can earn “Artifacts Available,” “Artifacts Evaluated,” and “Results Reproduced” badges, giving reviewers a concrete, checkable process.

SUMMARY

From Principle to Practice

Policy	Which ACM Code principle it operationalizes
Plagiarism policy	1.3 Be honest and trustworthy; 1.5 Respect the work of others
Authorship policy / CRediT	1.3 Honesty; 2.1 High quality of process, not just product
Conflict of interest policy	1.3 Honesty; 1.4 Fairness and non-discrimination
Peer review policy	1.7 Honor confidentiality; 2.4 Provide appropriate professional review
Reproducibility / artifact badges	2.5 Comprehensive, thorough evaluation of systems and their impacts

FURTHER READING

Student Reading List

ACM Code of Ethics and Professional Conduct	<i>ACM, 2018.</i>	The full text, 24 imperatives across 4 sections.
Susan Fowler, “Reflecting on One Very, Very Strange Year at Uber”	<i>susanjfowler.com, 2017.</i>	The original blog post, in her own words.
Matthew Prince, internal memo on the Daily Stormer decision	<i>Published via Gizmodo, Aug. 2017.</i>	The CEO's own account of an arbitrary content-moderation call.
Beauchamp & Childress, Principles of Biomedical Ethics	<i>Oxford University Press.</i>	The source for beneficence, non-maleficence, autonomy, justice.
The Russell–Einstein Manifesto	<i>Pugwash Conferences, 1955.</i>	Short, historic, and still directly readable.
Asilomar AI Principles	<i>Future of Life Institute, 2017.</i>	23 principles — compare directly against the 1975 Asilomar guidelines.
ACM Publications Policies: Plagiarism, Authorship, COI, Peer Review	<i>ACM.org/publications/policies.</i>	The operational rules behind today's research-integrity slides.